The Advantage Spine: Dynamic Coordinate Selection for High-Rate RLVR

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Abstract

Large learning rates can accelerate reinforcement learning with verifiable rewards (RLVR), yet dense updates deteriorate as the rate rises. We ask whether this degradation is jointly constrained by negative-advantage exposure and optimizer-update support. Positive- and negative-advantage updates exhibit a conflict-enriched heavy tail, providing a candidate explanation for high-rate degradation. Based on this observation, we introduce the *Advantage Spine*, a dynamic support that retains the per-tensor magnitude top 40% of the pre-mask AdamW update without pruning weights or changing the inference network. The Spine recipe combines this dynamic support with negative-advantage down-weighting. In a three-seed $2\times2\times4$ factorial on Qwen3-8B, neither component alone recovers the dense 1× reference at 20×. The Spine recipe reaches 0.445 Mean4, a paired gain of 0.019 over dense 1×, and remains above an exploratory lightly tuned dense 20× arm at 0.383. Its 39.3% support retains 85.0% of update energy and exhibits seed-specific agreement beyond mismatched Spine–dense and dense–dense controls. Density-matched mask controls further support non-arbitrary selection, while directionally consistent transfers under DAPO and on Qwen3-1.7B and Llama3.1-8B indicate limited portability. These results identify a reproducible, protocol-dependent high-rate operating region shaped jointly by advantage exposure and dynamic coordinate selection.

Keywords: reinforcement learning with verifiable rewards, sparse optimization, learning-rate stability, coordinate selection, large language models

1. Introduction

RLVR is an important post-training route for improving mathematical reasoning in large language models [1, 2], but its gains depend on optimization stability. Raising the learning rate accelerates parameter movement while applying the same global multiplier to every dense update coordinate. In our Qwen3-8B ladder, dense RLVR deteriorates as the rate rises. We therefore ask whether high-rate degradation depends not only on step size, but also on which optimizer-update coordinates are exposed to the negative-advantage signal.

We first examine the structure of the positive- and negative-advantage updates. Their active coordinates overlap at 7.4× an independence reference, and 52% of the shared coordinates have conflicting signs. This structure does not make negative advantages intrinsically harmful; it instead suggests that a uniformly amplified dense step can expose a conflict-enriched tail. Prior work shows that policy-gradient updates can form reproducible subnetworks [3, 4] and that sparse updates can preserve optimization progress [5, 6, 7]. What is missing is an online rule for selecting the RLVR update coordinates that receive an amplified step, together with a controlled test of how this support interacts with the negative-advantage channel.

Based on this observation, we introduce the Advantage Spine, which retains the largest 40% of the pre-mask AdamW update within each sufficiently large tensor and refreshes this support at every step. The Spine recipe combines the dynamic support with negative-advantage down-weighting. It selects optimizer-update coordinates without removing model weights or changing the inference network. We separate the two components and their rate dependence in a complete 16-cell factorial over mask presence, negative-channel weight, and four learning-rate multipliers, with three matched seeds per cell. Figure 1 summarizes the intervention and its principal evidence.

The evidence has three layers. At 20×, neither component alone recovers the dense 1× reference, whereas the joint Spine recipe reaches 0.445 Mean4 and remains above a lightly tuned dense arm. Its support concentrates update energy and aligns seed-specifically with dense-emergent coordinates; mismatched, dense–dense, and density-matched mask controls rule out random overlap or a shared dense backbone as sufficient explanations. The direction of effect also transfers, with smaller gains, under DAPO and on Qwen3-1.7B and Llama3.1-8B. Together, these results establish a reproducible, protocol-dependent high-rate operating region rather than a universal stability boundary.

2. Related work

Negative signals in RLVR.. GRPO and DAPO optimize verifiable reasoning rewards without the learned critic used in conventional PPO [8, 1, 2]. Negative-reward and negative-

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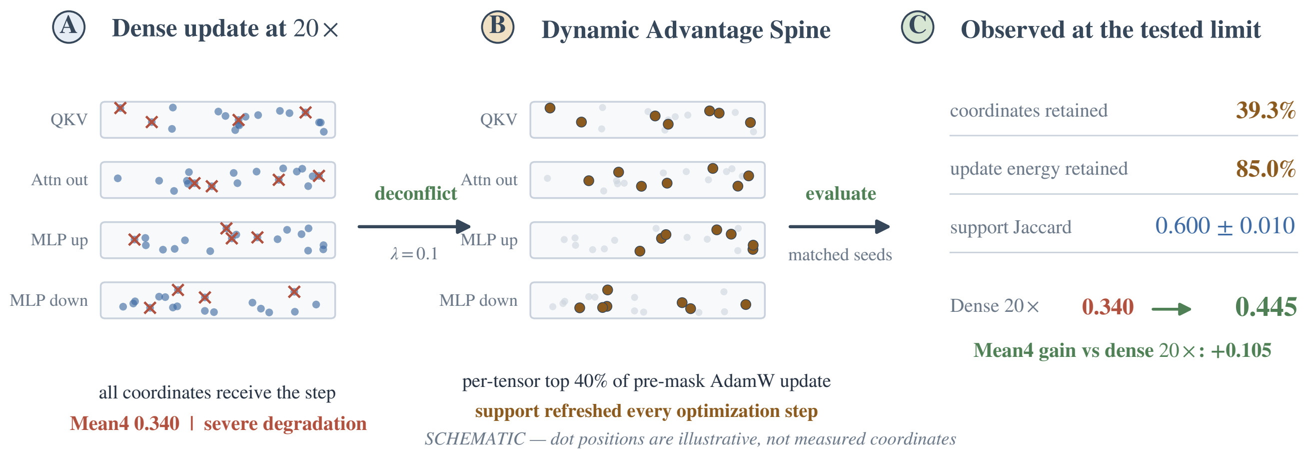

Figure 1: **From a dense high-rate update to the Spine recipe.** (A) A dense 20× update applies the optimizer step to every coordinate, including coordinates affected by competing advantage channels. (B) The Spine recipe down-weights negative-advantage policy-loss terms and dynamically retains the per-tensor top 40% of the pre-mask AdamW update. (C) At the largest tested rate: 39.3% nonzero coordinates retain 85.0% of update energy, support Jaccard with the matched dense-emergent support is 0.600 ± 0.010, and Mean4 increases from 0.340 to 0.445 relative to dense 20×. The matrix dots are a schematic only: their positions are illustrative and are not measured coordinates.

advantage samples are not uniformly helpful or harmful. Token-selective penalty attenuation can reduce Lazy Likelihood Displacement [9], whereas negative-sample reinforcement can improve the Pass@k spectrum [10]. These studies intervene at the token, sample, or objective level. We instead ask how a channel-wide coefficient interacts with optimizer-coordinate support when the global step is amplified.

Sparse updates and emergent subnetworks.. Lottery-ticket, dynamic sparse-training, and intrinsic-dimension results show that effective parameter movement can occupy a limited subspace or subnetwork [11, 12, 13, 14]. Gradient sparsification commonly uses error feedback [6, 15], while pruning removes or restructures persistent weights [16, 17, 18, 19]. Advantage Spine instead refreshes optimizer-update support at every step; the model and inference graph remain dense. Stable update subnetworks observed in RL fine-tuning [3] serve only as an offline identity check and are unavailable to the online recipe. This also differs from calibrating Adam's adaptive rate [20]: we hold the optimizer family fixed and vary which update coordinates receive the global multiplier.

RLVR update geometry.. Recent work characterizes RLVR through sparse critical tokens [21], update directions [22], low-rank dynamics [23], random sparse subnetworks [24], and optimizer-induced sparse movement [4]. Complementary analyses study learning outside principal directions [25] and step-size thresholds under a gradient-gap view [26]. Together, these results establish that RLVR updates are structured. They do not isolate the joint, rate-conditional interaction between negative-channel attenuation and online top-k optimizer-coordinate selection examined here.

Large-step stability.. Catapult and edge-of-stability dynamics show that neural optimization can remain stable beyond classical smoothness predictions [27, 28]. Our goal is narrower: we test an operational channel-and-coordinate intervention within a specified RLVR protocol, rather than propose a universal large-step mechanism.

3. Method

Our design objective is to restrict which advantage contribution and update coordinates are exposed to an amplified global step without changing the model architecture. Let $A_{t,i}$ be the group-standardized GRPO advantage and $\ell^{\text{clip}}_{t,i}$ the corresponding token-level policy loss after PPO ratio clipping and dual clipping. The implementation applies

$$\ell_{t,i}(\lambda) = \begin{cases} \ell^{\text{clip}}_{t,i}, & A_{t,i} \geq 0, \\ \lambda \ell^{\text{clip}}_{t,i}, & A_{t,i} < 0, \end{cases} \qquad \lambda \in \{0.1, 1.0\}. \tag{1}$$

The response-mask aggregation follows this scaling. Positive-advantage terms are unchanged, while the fixed KL penalty is added separately and is not scaled by λ. Thus $\lambda = 0.1$ attenuates, rather than deletes, the negative-advantage contribution; it does not modify rewards.

AdamW maps the resulting gradient and its moment state to a pre-mask update u_t. For every tensor with at least 4096 elements, M_t keeps the largest fraction $\tau = 0.4$ by $|u_t|$; smaller tensors remain dense. Selection uses the backend `torch.topk` implementation, and the realized supports are recorded by the mask-identity manifests. The applied update is

$$\theta_{t+1} = \theta_t - \eta M_t \odot u_t. \tag{2}$$

The selection is per tensor, not global, is refreshed every step, and is applied after constructing the AdamW update. There is no error feedback. We call the dynamic support $S_t = \text{supp}(M_t)$ the Advantage Spine and the joint condition $(\tau, \lambda) = (0.4, 0.1)$ the Spine recipe. SparseAdamW updates both Adam moment buffers densely, optionally applies its adaptive update-RMS clip to the bias-corrected Adam direction, masks that direction only in eligible tensors, and applies decoupled weight decay densely. The coordinate mask therefore does not mask moment updates or decoupled weight decay.

Three objects remain distinct. M_t is the online mask; its bit-packed bitmap and hash record its recoverable identity. S_{emg} is an offline reference built from the largest cumulative parameter movements of a density-matched dense 1× run and is never used by the training recipe.

We measure concentration as

$$E_t = \frac{\|M_t \odot u_t\|_2^2}{\|u_t\|_2^2}, \tag{3}$$

and support agreement with Jaccard $J = |S_t \cap S_{\text{emg}}|/|S_t \cup S_{\text{emg}}|$, recall $|S_t \cap S_{\text{emg}}|/|S_{\text{emg}}|$, and sign agreement on their intersection. For two independent sets of density 0.4, expected Jaccard is $0.4/(2-0.4) = 0.25$. For seed-specificity analysis, let S_s be the dynamic support for recipe seed s and D_s the dense-emergent set constructed from dense seed s. We report all nine $J(S_s, D_t)$ values, the three off-diagonal $J(D_s, D_t)$ values, and the per-seed specificity gain

$$G_s = J(S_s, D_s) - \frac{1}{2}\sum_{t \neq s} J(S_s, D_t). \tag{4}$$

These comparisons separate matched alignment from task-wide coordinates that recur across runs. Because sets are reused across pairs, their sample deviations are descriptive rather than independent-replicate inferential errors.

4. Experimental protocol

Primary factorial.. The primary experiment uses Qwen3-8B and a GRPO objective. All runs use model-only initialization from the same math-supervised SFT checkpoint; this is not an RL resume. The RL state is initialized at step 0, and every run completes 160 RL optimizer steps. We cross mask $\in$ {dense, top-0.4}, negative-channel coefficient $\lambda \in \{1.0, 0.1\}$, and learning-rate multiplier $\in \{1, 5, 10, 20\}$. The 1× rate is 5×10^{-6}. Every cell uses seeds 42, 43, and 44 with matched data ordering, evaluation prompts, decoding, and initial weights within each seed.

Evaluation protocol.. We evaluate Math500 ($n = 500$), AIME24 and AIME25 (30 problems each), and the English split of OlympiadBench ($n = 581$). Each checkpoint is evaluated with avg@32 accuracy at temperature 0.6, top-p = 1.0, no top-k truncation, and an 8192-token response cap. We evaluate RL steps 40, 80, 120, and 160 and use step 160 for the primary comparisons. The resulting decoded-sample counts are $30 \times 32 = 960$ per AIME set, 16,000 for Math500, and 18,592 for OlympiadBench. Mean4 is the unweighted arithmetic mean of the four benchmark accuracies. Training seeds index independent RL runs; generations per problem are evaluation replicates, not additional training seeds.

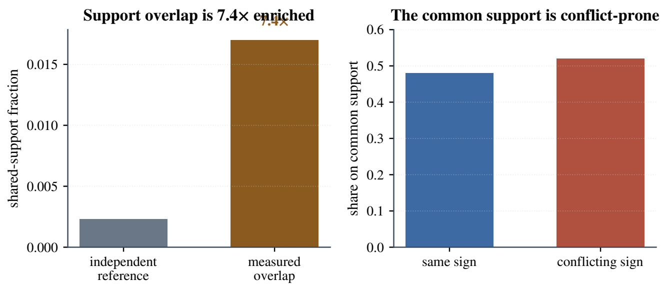

Figure 2: **Positive and negative channels occupy overlapping, conflict-prone coordinates.** The 1.7% overlap is 7.4× an independence reference; signs conflict on 52% of the shared support. Values are three-seed means; enrichment is the ratio of aggregate quantities.

Initialization, recovery, and learning rates.. New runs load only SFT model weights. Full-state recovery is restricted to the same model, factorial cell, seed, and output directory. Requested, optimizer, scheduler, and logged learning rates agree for every included run; an exploratory Llama continuation with a logged-rate mismatch is excluded. Learning-rate and state verification appears in Appendix A; the complete initialization, recovery, and evaluation contract appears in Appendix B.

Uncertainty and estimands.. Factorial cells report mean and sample standard deviation across three seeds. After reporting the complete ladder, we highlight the endpoint contrast between dense 1× and recipe 20×, paired within seed. Its two-sided Student-t interval is a nominal description of variation across these matched seeds, not a selection-adjusted confirmatory interval over the 16 factorial cells. Table 4 therefore reports all 48 seed-level scores. For the Qwen3-1.7B transfer, all four benchmark components are available for three matched seeds. In addition to Mean4, we report a small-set sensitivity summary, Mean2, as the average of Math500 and Olympiad after excluding AIME24 and AIME25. Mask-control and aggregate-only transfer entries are identified wherever per-seed uncertainty is unavailable. The lightly tuned dense 20× arm is likewise an aggregate diagnostic without matched-seed uncertainty and is not treated as a factorial cell.

5. Results

5.1. The two advantage channels share a conflict-prone tail

Across three seeds, positive and negative channels are each active on 4.8% of coordinates. Their measured shared support is 1.7%, compared with $0.048^2 = 0.23\%$ under an independence reference, a 7.4× enrichment. On the shared support, 48% of signs agree and 52% conflict. The positive and negative update distributions are both tail-concentrated (Gini 0.70 and 0.63; top-1% energy 0.88 and 0.84). This conflict-enriched tail motivates the intervention as a candidate mechanism for step-size sensitivity; it does not imply that useful negative-sample reinforcement should be removed.

Table 1: Primary Qwen3-8B benchmark decomposition at RL step 160, aggregated over three matched seeds. Benchmark columns and Mean4 are independently rounded to three decimals.

Condition	Math500	AIME24	AIME25	Olympiad	Mean4
Dense 1×	.762	.227	.200	.513	.426
Recipe 20×	.800	.240	.210	.530	.445

5.2. The complete factorial isolates the joint high-rate effect

Table 4 and Fig. 3 contains every factorial cell. Dense $\lambda = 1$ declines slightly at 5× and sharply thereafter, reaching 0.340 at 20× (−0.086 from its 1× value). Down-weighting the negative channel without masking reaches 0.400 at 20×; masking without down-weighting reaches 0.410. Both improve over dense 20×, but neither recovers the dense 1× reference of 0.426.

The joint recipe instead rises from 0.426 at 1× to 0.445 at 20×. At the endpoint, its gain is 0.105 over dense 20× and 0.019 over dense 1×. The three seed-paired gains over dense 1× are 0.020, 0.016, and 0.021, giving a nominal 95% paired interval $[0.0124, 0.0256]$.

The recipe improves every benchmark component, including Math500 and Olympiad; its Mean4 gain is therefore not driven solely by the two smaller AIME sets.

At 20×, the conventional 2×2 interaction is $0.445-0.410-0.400+0.340 = -0.025$. The result is therefore operational complementarity, not positive synergy: both components are required for the only endpoint condition that exceeds the native dense reference. The tested ladder ends at 20× and does not locate a failure boundary for the recipe.

An exploratory lightly tuned dense 20× arm (16 warmup steps, global gradient clip 0.5, and $\beta_1 = 0.8$) recovers Mean4 from 0.340 to 0.383, showing that part of the untuned degradation is optimizer-schedule sensitive. It ends with reward −0.08, KL loss 0.04–0.07, entropy 0.18–0.23, response clip ratio 0.25–0.45, and no recorded run failure, numerical collapse, or policy collapse; we classify it as degraded but partially recovered. The endpoint nevertheless remains below dense 1× (0.426) and the Spine recipe at 20× (0.445; gap 0.062). Because this arm is an aggregate diagnostic rather than a matched-seed tuning study, it establishes partial schedule sensitivity, not superiority over tuned dense optimization. The Spine recipe therefore expands the high-rate operating region under the tested protocol without implying superiority over every retuned dense optimizer.

Table 2 shows why the full LR ladder matters. At native rate neither component helps materially. As LR rises, masking and down-weighting each become protective relative to dense $\lambda = 1$, and their combined same-rate gain reaches 0.105 at 20×. The interaction is small and changes sign across the ladder; the gain is therefore a rate-dependent joint operating condition, not a stable super-additive effect.

This rate dependence reconciles the result with evidence that negative reinforcement can be useful [9, 10]. In the dense arm, lowering λ changes Mean4 by only +0.002 at 1× but by +0.060 at 20×. Attenuation is therefore protective when a large global multiplier exposes the conflict-prone tail. The conclusion is step-size–conditional: we do not infer that negative samples should generally be suppressed.

Table 2: Factorial contrasts at each LR. Mask effect is Top-0.4 minus Dense at $\lambda = 1$; down-weight effect is $\lambda = 0.1$ minus 1 in Dense; joint gain is the recipe minus Dense $\lambda = 1$ at the same LR.

LR	Mask	Down-weight	Joint gain	Interaction
1×	−.005	.002	.000	.003
5×	.000	.015	.014	−.001
10×	.018	.015	.045	.012
20×	.070	.060	.105	−.025

Table 3: Endpoint stability diagnostics at 20×. Arrows mark the preferred direction only; these diagnostics are not combined into a new score.

Condition	Reward ↑	KL	Entropy	Clip ↓	Classification
Dense	−.150	.090	.150	.650	degraded
Spine recipe	.080	.030	.250	.100	normal

The optimization diagnostics agree with the endpoint accuracy rather than only its aggregate score. At 20×, the recipe ends with reward 0.080, KL loss 0.030, entropy 0.250, and response clip ratio 0.100. Dense 20× ends with reward −0.150, KL loss 0.090, entropy 0.150, and response clip ratio 0.650. Neither endpoint meets the operational criteria used in this paper for a run failure, numerical collapse, or policy collapse; Dense 20× is classified as severe performance degradation. The component-level endpoint results are reported in Table 1.

5.3. The online support is concentrated and non-arbitrary

The factorial establishes the performance pattern. We next test whether the selected support carries specific structure or merely behaves as an arbitrary density-matched sparsification.

The realized nonzero ratio is 0.393 rather than exactly 0.400 because small tensors are exempt and top-k counts are discrete. This support retains 0.850 of pre-mask update energy, corresponding to a retained norm ratio of 0.922. This establishes concentration without assuming that the discarded 15% is harmless.

The identity analysis uses source-attested step-160 mask manifests and hash records. The public artifact verifies their identities, protocol fields, and package integrity; the remote raw mask files are not redistributed.

Across seeds, the step-160 dynamic support agrees with its matched dense- emergent set: Jaccard 0.600 ± 0.010, recall 0.750 ± 0.008, sign agreement 0.780 ± 0.005, and enrichment 2.400 ± 0.039 over the random reference. Because dense trajectories can share task- and initialization-dependent heavy coordinates, we test structured rather than only random nulls. The complete cross-seed matrix has matched diagonal $0.604, 0.589, 0.607$, whereas all six mismatched Spine–dense values lie between 0.409 and 0.431 (0.420 ± 0.008). The three dense–dense values are $0.414, 0.423, 0.419$ (0.419 ± 0.005). Thus matched alignment exceeds the dense–dense baseline by 0.181 and the mismatched baseline by 0.180; after removing the dense–dense shared component, the normalized specificity is $(0.600 - 0.419)/(1 - 0.419) = 0.312$.

The row-wise specificity gains G_s are $0.191, 0.167, 0.182$, or 0.180 ± 0.012. Every matched value exceeds every mis-

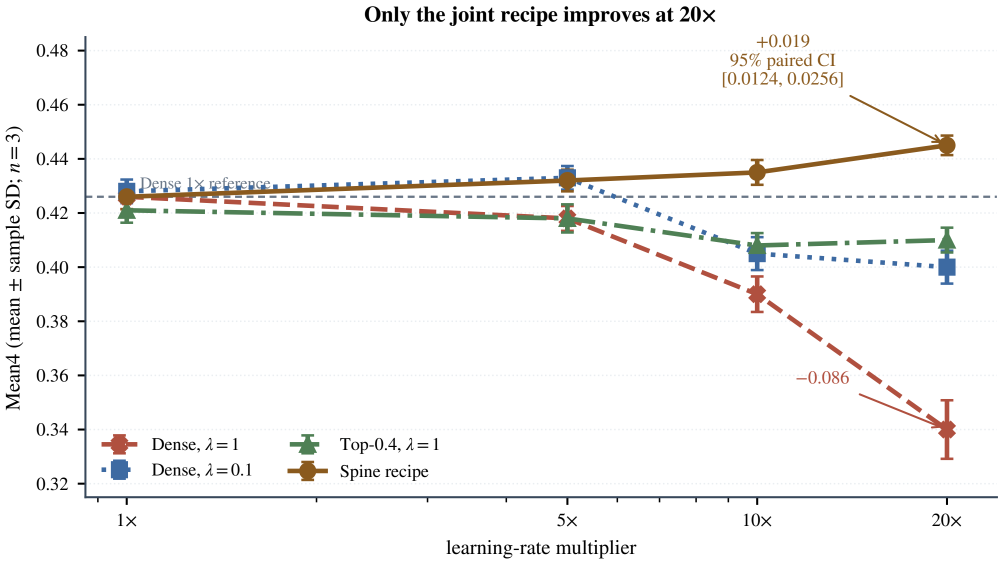

Figure 3: **Three-seed factorial.** Mean ± sample SD in every cell. Dense training degrades as LR rises; only the combined dynamic-mask, $\lambda = 0.1$ recipe improves monotonically over the tested ladder.

Table 4: Complete Qwen3-8B factorial. Each cell lists seeds 42/43/44 followed by mean ± sample SD; logged and requested learning rates agree.

Mask	λ	LR	s42	s43	s44	Mean±SD
Dense	1.0	1	.421	.430	.427	.426±.0046
		5	.423	.414	.417	.418±.0046
		10	.384	.397	.389	.390±.0066
		20	.352	.331	.337	.340±.0108
Dense	0.1	1	.431	.423	.430	.428±.0044
		5	.428	.436	.435	.433±.0044
		10	.409	.398	.408	.405±.0061
		20	.404	.393	.403	.400±.0061
Top-0.4	1.0	1	.417	.426	.420	.421±.0046
		5	.421	.412	.421	.418±.0052
		10	.404	.413	.407	.408±.0046
		20	.406	.415	.409	.410±.0046
Top-0.4	0.1	1	.430	.421	.427	.426±.0046
		5	.428	.436	.432	.432±.0040
		10	.439	.430	.436	.435±.0046
		20	**.441**	**.446**	**.448**	**.445±.0036**

matched or dense–dense value. This clean diagonal separation supports seed-specific alignment beyond a common dense backbone. The three-seed identity result is descriptive; Jaccard and recall are reported for comparability and are algebraically linked for equal-size sets.

The mask controls provide a stronger test than overlap alone (Fig. 4D). At 20×, the dynamic-support recipe obtains 0.445 Mean4, compared with 0.430 for a frozen support, 0.440 for the offline dense-emergent oracle, 0.420 for a static magnitude mask, 0.400 for a random 40% mask, and 0.350 for the reported complementary mask. These aggregate controls show that dynamic selection approaches the offline oracle numerically and substantially exceeds arbitrary or complementary supports; the 0.005 oracle difference has no per-seed uncertainty.

Table 5: Transfer summary. J denotes matched support Jaccard. The primary Qwen3-8B GRPO, Qwen3-1.7B, and matched Llama comparisons have three seeds; DAPO is an aggregate result.

Setting	J	Dense	Recipe	Gain
Qwen3-8B GRPO	.600	.426	.445	+.019
Qwen3-8B DAPO	.600	.440	.455	+.015
Qwen3-1.7B	.580	.364	.380	+.016
Llama3.1-8B	.580	.169	.179	+.010

5.4. *The direction transfers across algorithms and models*

The pattern transfers with smaller absolute gains (Table 5). Under DAPO on Qwen3-8B, Mean4 changes from 0.440 to 0.455. On Qwen3-1.7B, the three-seed Mean4 changes from 0.36413 ± 0.00646 to 0.38037 ± 0.00760 (paired gain 0.01624 ± 0.00482). The transfer is not wholly carried by the two small AIME sets: Mean2 changes from 0.60257 ± 0.00592 to 0.61108 ± 0.00714, a paired gain of 0.00851 ± 0.00628, positive in all three seeds. The component result is mixed rather than universal: Math500 changes by −0.0013 while Olympiad changes by +0.0184. On a matched Llama3.1-8B rerun Mean4 changes from 0.169 to 0.179; all three recipe seeds avoid collapse, with final logged LR 10^{-4}, reward 0.020, KL loss 0.030, entropy 0.220, and response clip ratio 0.180. The Llama ladder saturates at 10–20× (both 0.179), unlike the monotonic Qwen3-8B ladder. The claim is therefore transfer of direction, not an invariant optimal multiplier.

5.5. *Training overhead is modest and inference remains unchanged*

On the 8× NVIDIA H200 setup used for these experiments (141 GiB per GPU), median RL-step time is 12.007 s for Dense and 13.207 s for the recipe; mean peak allocated memory per GPU is 72.25 and 75.86 GiB. Thus mask construction

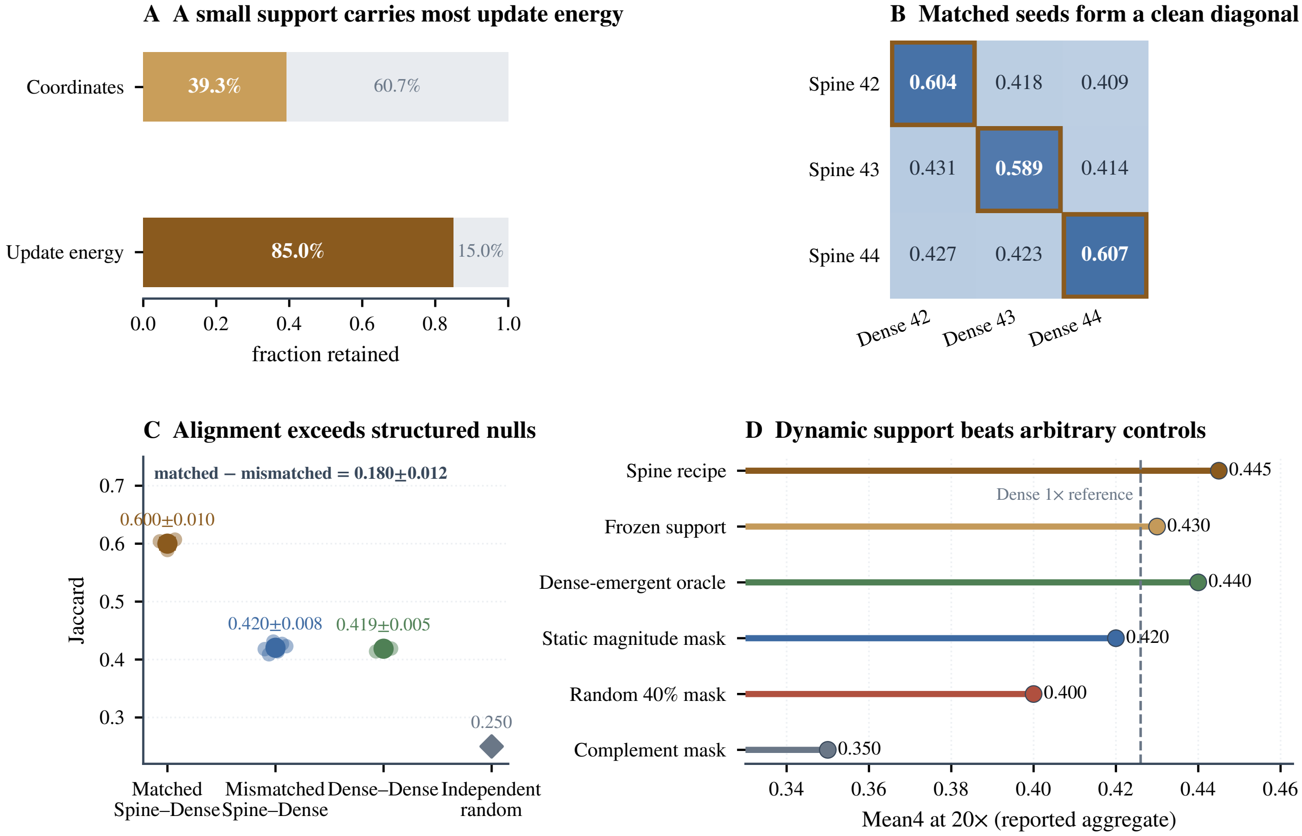

Figure 4: **The selected support is concentrated, seed-specific, and non-arbitrary.** (A) The realized 39.3% support retains 85.0% of squared update norm. (B) The complete Spine–dense cross-seed matrix has a clean matched diagonal. (C) Matched alignment exceeds both mismatched Spine–dense and dense–dense structured controls, not only an equal-density random reference; error bars are descriptive because sets are reused across pairwise comparisons. (D) At matched reported density and 20×, dynamic support matches the offline oracle numerically and exceeds frozen, static, random, and complementary controls; these mask-control values are aggregate results without per-seed uncertainty.

Table 6: Qwen3-1.7B benchmark sensitivity across three matched seeds, recomputed from integer correct/total counts. Entries are mean ± sample SD; Mean2 excludes AIME24/25 and averages Math500 with Olympiad. The final row is the paired mean difference.

Condition	Math500	AIME24	AIME25	Olympiad
Dense	.742 ± .006	.119 ± .008	.132 ± .007	.463 ± .006
Recipe	.741 ± .008	.151 ± .009	.149 ± .007	.481 ± .006
Δ	−.001	+.031	+.017	+.018

Condition	Mean4	Mean2
Dense	.36413 ± .00646	.60257 ± .00592
Recipe	.38037 ± .00760	.61108 ± .00714
Δ	+.01624	+.00851

adds 10% step time and 5% peak allocated memory in this implementation. It changes no inference graph and has zero inference overhead once training is complete. Across all 81 runs, the records contain 1,332.65 node-hours, or 10,661.2 H200-GPU-hours; this end-to-end total includes evaluation and checkpointing and is not a pure training-throughput measure. These measurements support feasibility but are not a hardware-wide benchmark.

6. Discussion

The factorial and controls support a conditional empirical picture. Down-weighting reduces exposure to a conflict-enriched contribution, while dynamic top-k avoids applying the amplified global step uniformly. Together they expand the high-rate operating region under the tested protocol. Concentrated energy, seed-specific alignment, and density-matched mask controls argue against arbitrary 40% selection as a sufficient explanation, while leaving curvature control, optimality, and the unique causal mechanism unresolved.

Dynamic online selection also differs from an offline oracle. It requires no completed dense trajectory, yet approaches the oracle aggregate and exceeds frozen or static supports. This supports adaptation of optimizer-update coordinates rather than persistent pruning, consistent with the broader role of changing support in dynamic sparse training [12, 29].

The negative-channel result is likewise conditional. Negative samples can carry useful learning signal [10], while uniform penalties can cause likelihood displacement [9]. Our factorial adds optimizer-scale exposure: attenuation has little effect at the native rate and becomes protective at the largest rate. The result therefore positions the conflict-prone coordinate tail as a constraint on usable step size, rather than negative rein-

Table A.7: Effective learning rates. Every column agrees within each row.

Label	Requested	Optimizer	Scheduler	Logged
1×	.000005	.000005	.000005	.000005
5×	.000025	.000025	.000025	.000025
10×	.000050	.000050	.000050	.000050
20×	.000100	.000100	.000100	.000100

forcement as a generally harmful objective.

Limitations.. The evidence is limited to three seeds, four mask snapshots over 160 RL steps, math RLVR, two model families, and finite dense retuning. The endpoint interval covers only matched seed variation and is not selection-adjusted. The ladder ends at 20×, and four mask snapshots cannot reveal a later stability boundary or continuous support churn. Transfer gains are modest and vary across benchmarks; the tuned dense arm also shows that part of the degradation is schedule-sensitive. Longer horizons, broader tasks, and more extensive dense retuning are needed to map the boundary and scope of the observed operating region; the present claim is not universal superiority over every retuned dense optimizer.

7. Conclusion

A 16-cell, three-seed factorial shows that negative-advantage exposure and dynamic optimizer-coordinate support jointly shape high-rate RLVR. The Spine recipe gains 0.019 Mean4 over dense 1× at the largest tested rate. Its support concentrates update energy, aligns seed-specifically with dense-emergent coordinates, and exceeds structured nulls and density-matched mask controls. These results identify a reproducible, protocol-dependent high-rate operating region and provide an operational tool for studying its coordinate structure.

Reproducibility statement

The accompanying artifact contains seed-level result tables, resolved configuration digests, and scripts that recompute every reported aggregate from those tables. Cross-cluster run and environment records are included as an author-attested export; local verification covers package structure, protocol fields, counts, and file integrity within the release, and does not remount or redistribute remote raw masks or checkpoints. Large checkpoints remain available from the corresponding author subject to storage and license constraints.

Appendix A. Learning-rate and state verification

Every new experiment begins at RL step 0 by loading only the designated SFT model weights. The parent SFT global step is provenance metadata and is not counted as an RL step. The optimizer and constant scheduler are initialized at step 0; RL checkpoints and evaluations are recorded at steps 40, 80, 120, and 160. LR assignment, optimizer parameter-group verification, scheduler verification, and JSONL verification yield zero mismatched runs or steps. Full-state recovery is reserved for an interruption of the identical run and restores optimizer, scheduler, trainer-extra, RL-step, data-position, and RNG states without changing the LR.

Appendix B. Implementation and evaluation contract

Training uses GRPO with 256 prompts per step and four responses per prompt (1,024 rollout sequences), a PPO mini-batch of 256, micro-batch size one per GPU, and eight GPUs. Prompt and response caps are 1,024 and 8,192 tokens. The objective uses symmetric clipping at 0.2, dual-clip $c = 3$, fixed in-loss low-variance KL with coefficient 0.001, zero entropy coefficient, group-mean advantage centering, and group-standard-deviation scaling. Dense updates use AdamW and Top-0.4 updates use SparseAdamW; both use $(\beta_1, \beta_2) = (0.9, 0.999)$, $\epsilon = 10^{-8}$, weight decay 0.01, global gradient clipping at 1.0, a constant LR schedule, and no warmup. Training rollouts sample four responses at temperature 1.0 and top-p = 1.0. Evaluation uses avg@32 at temperature 0.6 and top-p = 1.0, with the same 8,192-token response cap. The canonical machine- readable contract and its digest are included in `evaluation_and_training_protocol.json`.

The exploratory tuned-dense diagnostic changes only three optimizer settings: 16 warmup steps, global gradient clipping at 0.5, and $\beta_1 = 0.8$. Its Mean4 and endpoint diagnostic ranges are released as an aggregate record; matched seed-level uncertainty is unavailable, so it is not pooled with the factorial or used as a tuned-dense optimum.

Appendix C. Llama exploratory conflict record

The excluded D18 run requested 10^{-4} but logged 5×10^{-6} and ended at step 56 with reward -1, AIME accuracy 0, entropy 0.021, response clip ratio 1.0, and collapse. D18b and D18c logged 10^{-4} correctly but also collapsed (steps 63 and 200). These runs are retained as negative evidence and are not pooled with the matched three-seed rerun in Section 5.4.

CRediT authorship contribution statement

Shikai Li: Investigation, Writing–original draft. **Qingsong Cai:** Writing–review and editing. **Rui Shi:** Supervision, Writing–review and editing.

Declaration of competing interest

The authors declare no known competing financial interests or personal relationships that could have influenced the work reported in this paper.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used Cursor and OpenAI Codex (AI-assisted coding and writing tools) to improve English phrasing and manuscript organization, assist with LaTeX formatting, and support the development and review of plotting and consistency-checking code. The authors reviewed and edited all AI-assisted outputs, independently verified the references, data-derived quantities, and scientific claims, and take full responsibility for the content of the published article. These tools were not used to generate or alter experimental observations or to synthesize image content. All tables, the graphical abstract, and the manuscript figures were produced deterministically from author-verified data using conventional code; all AI-assisted code was reviewed and executed under author control.

Data availability

Result tables, analysis scripts, figure generators, run identities, resolved configurations, artifact digests, and the Advantage Spine implementation are included in the accompanying open repository `https://github.com/vivia-an/advantage_spine_v1`. Qwen3, DAPO-Math-17K, AIME25, and VERL are used under Apache-2.0; Llama-3.1 uses its Community License and Acceptable Use Policy; OlympiadBench uses MIT. Because the referenced MATH500 and AIME24 cards do not declare a license, the artifact releases their hashes, identities, counts, and aggregate results but not their problem text. The public mask evidence likewise consists of verified identity, protocol, and hash records rather than the remote raw mask files. The complete resource matrix is in `LICENSE_AND_DATA_USE.md`. Checkpoint redistribution remains subject to all upstream model and data terms.

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